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MTGeophysics.jl: A software repository for magnetotelluric research and applications

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Software

- Review
- Repository
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Summary

MTGeophysics.jl is a Julia package for magnetotelluric (MT) geophysical modelling, inversion, and interactive visualization. The magnetotelluric method uses natural electromagnetic signals to image the electrical resistivity structure of the Earth's subsurface, with applications ranging from mineral exploration to studies of tectonic structure. MTGeophysics.jl provides 1D and 2D forward solvers, stochastic inversion based on Very Fast Simulated Annealing (VFSA) (Sen & Stoffa, 2013) in 2D and 3D, I/O routines for the widely used ModEM 3D inversion code (Kelbert et al., 2014), and interactive 3D model viewers built on GLMakie. Because many different subsurface configurations can explain observed MT responses, the VFSA workflow generates an ensemble of plausible models rather than a single deterministic solution, enabling quantitative uncertainty assessment. The package is designed to be accessible to researchers who need an integrated, scriptable environment for MT data processing and interpretation within the Julia ecosystem.

Statement of need

Magnetotelluric surveys produce large volumes of multi-frequency, multi-station electromagnetic data that must be processed through forward modelling and inversion to obtain subsurface resistivity images. Existing MT software is predominantly written in Fortran, MATLAB, or Python: ModEM (Egbert & Kelbert, 2012; Kelbert et al., 2014) is a widely used 3D inversion code written in Fortran; MARE2DEM (Key, 2016) handles 2D/3D marine electromagnetic modelling in Fortran; MTpy (Kirkby et al., 2019) is a Python toolbox for MT data processing and visualization; and numerous MATLAB toolboxes exist for 1D and 2D analysis. However, these tools are often disconnected, with separate programs for forward modelling, inversion, file conversion, and visualization, requiring researchers to maintain ad-hoc scripts in multiple languages to bridge them.

MTGeophysics.jl addresses this fragmentation by providing a single Julia package that spans the MT modelling and interpretation workflow, from forward modelling and inversion to a final preferred subsurface model with quantified uncertainty. The package is designed as a research repository: its forward solvers, data structures, and inversion routines serve as reusable building blocks so that implementing new ideas (alternative parameterisations, hybrid inversion strategies, or novel uncertainty quantification schemes) is straightforward without rebuilding core MT tooling from scratch. Because Julia's type system and multiple dispatch make it straightforward to compose packages, MTGeophysics.jl is designed as a core component of a broader JuliaGeophysics ecosystem. Julia's composability with automatic differentiation and scientific computing libraries makes the package a natural foundation for extending classical MT workflows with modern numerical methods and scientific machine learning.

The package targets MT researchers and students who want a unified, open-source toolkit that builds on Julia's (Bezanson et al., 2017) strengths in numerical computing, automatic differentiation, composability, and interactive graphics. It reads and writes standard ModEM file formats, enabling direct interoperability with the Fortran ModEM code used by many research groups worldwide.

State of the field

Several open-source tools address parts of the MT workflow. ModEM (Egbert & Kelbert, 2012; Kelbert et al., 2014) is the community standard for 3D MT inversion but provides no built-in visualization or 1D/2D forward capabilities. MARE2DEM (Key, 2016) focuses on 2D and 3D marine electromagnetic forward and inverse modelling but is Fortran-based and limited to controlled-source methods. MTpy (Kirkby et al., 2019) provides comprehensive Python utilities for MT data handling, processing, and visualization but does not include forward solvers or stochastic inversion. pyGIMLi (Rücker et al., 2017) and SimPEG (Cockett et al., 2015) are Python frameworks for general geophysical inversion that include MT modules, but their MT-specific functionality is embedded within much larger codebases. None of these packages offer integrated stochastic inversion with ensemble uncertainty quantification in both 2D and 3D.

MTGeophysics.jl contributes a tightly integrated Julia-native package that covers 1D and 2D MT forward modelling, 2D and 3D VFSA inversion with ensemble uncertainty quantification, and interactive 3D model viewing with GIS overlay support. The choice of Julia provides performance comparable to compiled languages while maintaining the interactivity and rapid prototyping advantages of dynamic languages.

Software design

MTGeophysics.jl is structured as a single Julia module with clearly separated functional layers. The core layer handles ModEM 3D data and model I/O, supporting the standard ModEM file formats for models (WinGLink/WS format) and impedance data. The 1D module implements both analytical recursive impedance solutions and finite-difference solvers on geometrically graded meshes. The 2D module constructs tensor meshes, assembles sparse finite-difference operators for the TE and TM mode Maxwell equations, and solves them using direct sparse factorization via LinearSolve.jl.

The VFSA inversion module (Sen & Stoffa, 2013) uses radial basis function (RBF) parameterization to map a reduced set of control points to the full model grid, enabling efficient stochastic search in a lower-dimensional space. In 2D the package includes its own finite-difference forward engine; in 3D the inversion wraps the ModEM forward solver (Kelbert et al., 2014), ensuring compatibility with established workflows. Multiple independent Markov chains run in parallel, and ensemble statistics (mean, median, standard deviation) are computed across chains to provide uncertainty estimates.

Interactive visualization is handled through GLMakie, providing GPU-accelerated 3D slice viewers (XY, XZ, YZ, and combined) with slider controls, coordinate reprojection via Proj.jl, and optional shapefile overlays for geological maps, faults, and survey boundaries. The package also supports exporting depth slices and cross-sections as georeferenced shapefiles suitable for integration into GIS platforms, closing the loop between geophysical modelling and geological mapping. The visualization layer is conditionally loaded, ensuring the core package functions without OpenGL dependencies.

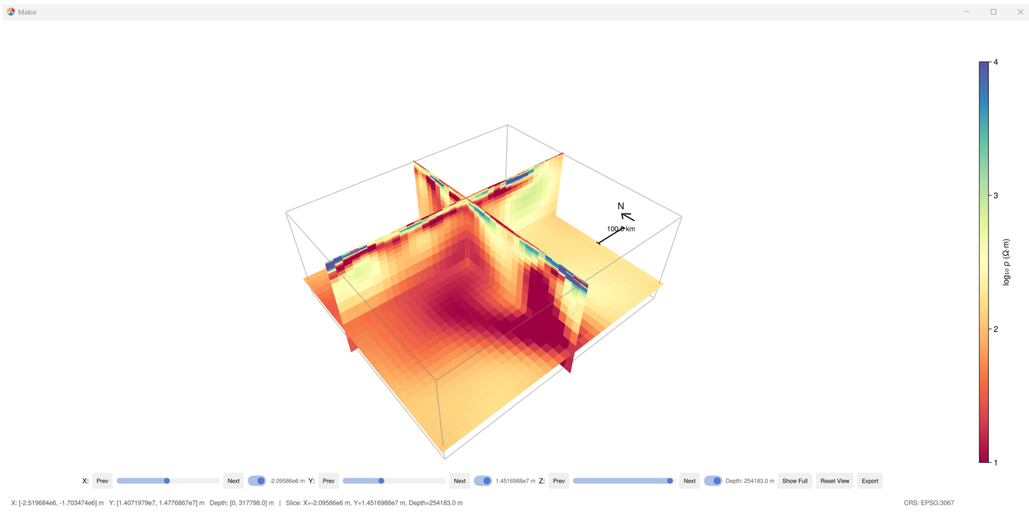

Figure 1: Interactive 3D model viewer showing combined XY, XZ, and YZ resistivity slices with slider controls, coordinate reprojection (EPSG:3067), north arrow, and scale bar. Depth, X, and Y sliders allow real-time browsing; the view can be toggled between core-only and full-padding extents.

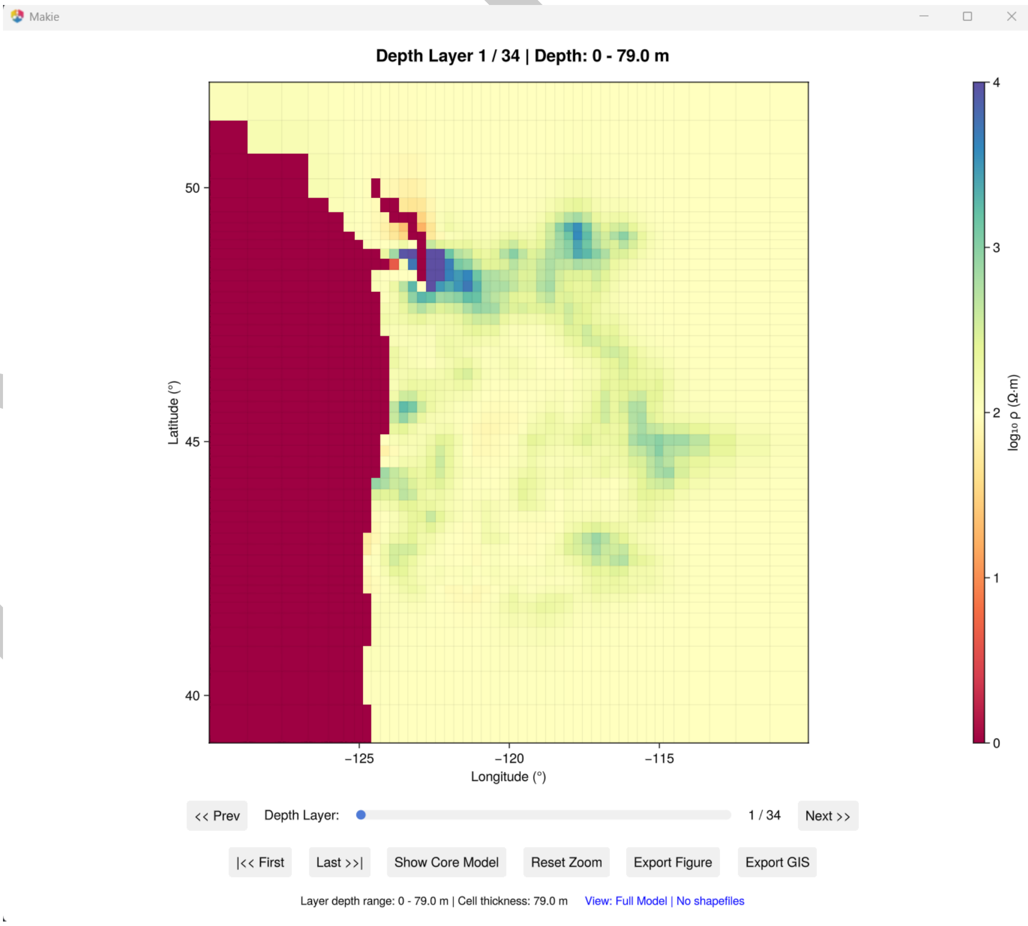

Figure 2: Interactive XY depth-slice viewer with WGS 84 geographic coordinates (EPSG:4326), depth-layer slider, core/full model toggle, and GIS shapefile overlay support. Slices can be exported as high-resolution PNGs or georeferenced shapefiles for direct import into GIS platforms.

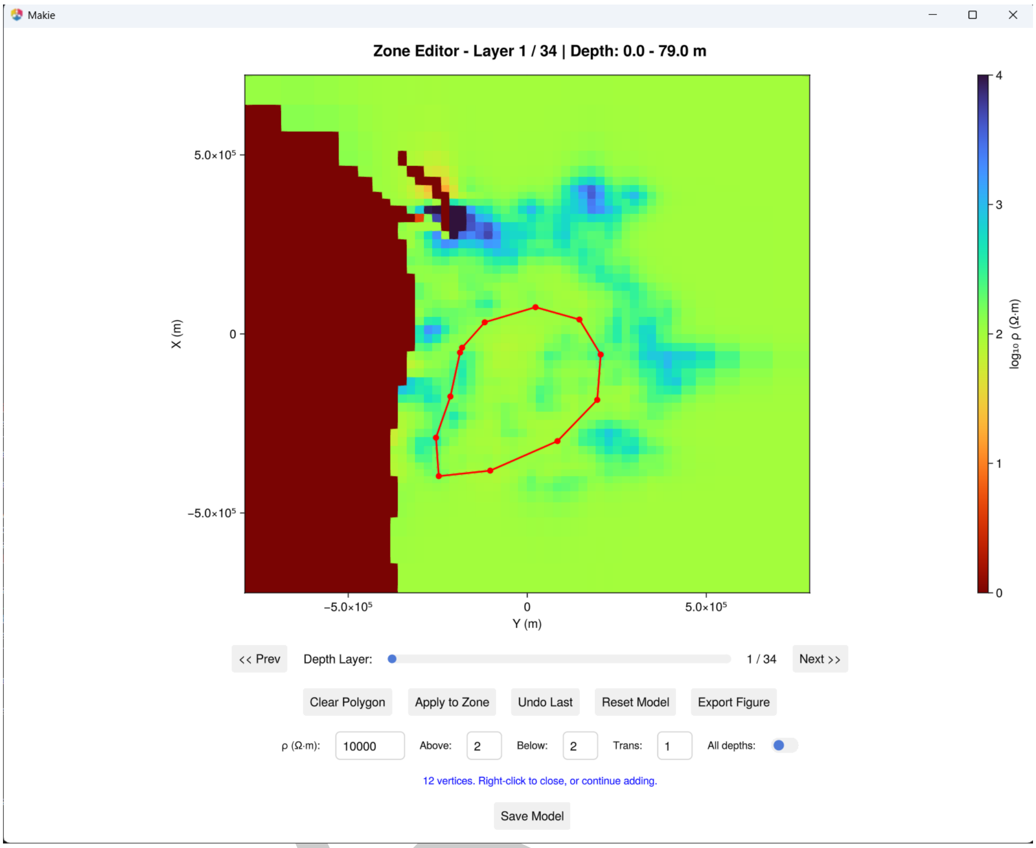

Figure 3: Polygon-based interactive model editor. The user draws a closed polygon on a depth slice, sets a target resistivity and vertical extent (layers above, below, and transition), and applies the edit. Undo, reset, and save controls allow iterative refinement of the 3D model before re-running the forward solver.

Publication-quality static plots use CairoMakie for data maps, response curves, model cross-sections, and convergence diagnostics. All file formats use plain text, ensuring reproducibility and version-control friendliness.

Benchmarks and validation

The package includes two levels of numerical validation, both fully reproducible from the repository.

COMMEMI 2D benchmarks. The 2D forward solver and VFSA inversion have been validated against the COMMEMI benchmark suite (Zhdanov et al., 1997), the community-standard test set for 2D MT numerical methods. The repository natively generates the COMMEMI-2D-I, II, and III synthetic models and observed data via the included benchmark scripts (`helpers/benchmarks_2d.jl`), so validation can be repeated with a single command without any external data downloads.

Cascadia 3D field-data benchmark. The 3D VFSA inversion has been benchmarked at regional scale using USArray MT data from the Cascadia subduction zone (Patro & Egbert, 2008), a dataset extensively studied with the deterministic ModEM NLCG inversion. This benchmark will be presented at EGU General Assembly 2026 (Mishra et al., 2026). Figure 4 compares depth slices from the published ModEM NLCG result (top row) with the VFSA ensemble mean computed from nine independent Markov chains (middle row), each running 3000 iterations with four trial proposals per iteration (approximately 60 seconds per forward solve). The VFSA ensemble mean recovers the major conductive structures identified by deterministic inversion at both shallow (24–30 km) and deeper (59–74 km) depth ranges, and in several areas produces sharper geological boundaries than the smoothness-regularised NLCG result. The bottom row shows the ensemble standard deviation, a per-voxel uncertainty estimate that no single deterministic inversion can provide. Regions of high standard deviation correspond to areas where the data poorly constrain the model, giving interpreters direct information about which features are robust and which remain ambiguous. Because the stochastic workflow produces a distribution of plausible models rather than one "best" solution, it quantifies the non-uniqueness inherent in MT inversion.

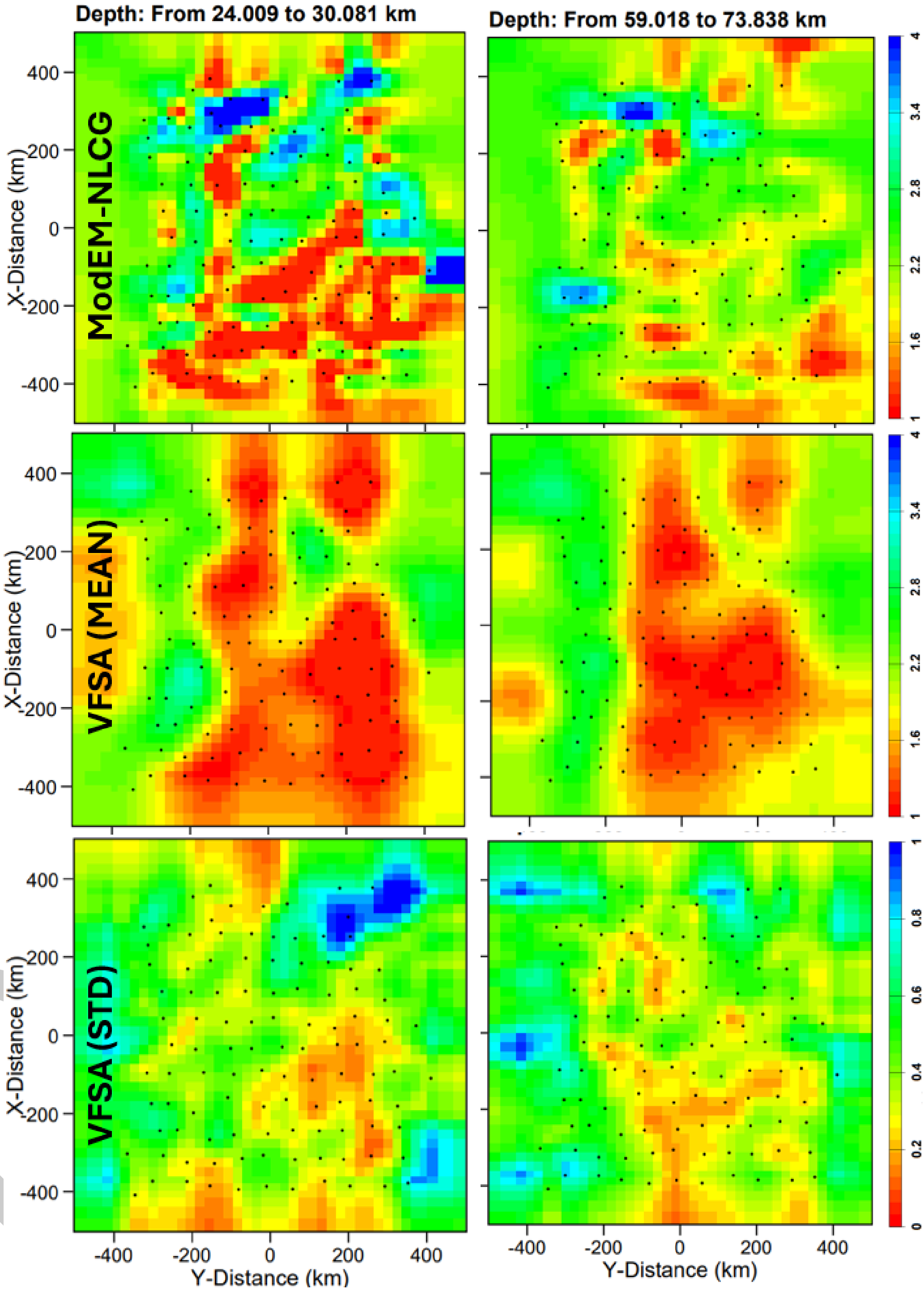

Figure 4: 3D VFSA benchmark on Cascadia field data. Top: ModEM NLCG deterministic inversion. Middle: VFSA ensemble mean from 9 independent chains. Bottom: ensemble standard deviation (uncertainty). Left column: 24–30 km depth. Right column: 59–74 km depth. The VFSA mean recovers the same major conductive features as the deterministic result while additionally providing spatially resolved uncertainty estimates.

Research impact statement

MTGeophysics.jl has been developed to support ongoing magnetotelluric research at the Geological Survey of Finland (GTK). The numerical core has been validated at two levels: the 2D forward solver and VFSA inversion reproduce the community-standard COMMEMI benchmark models (Zhdanov et al., 1997), and the 3D VFSA inversion recovers established resistivity structures from the published Cascadia USArray dataset (Patro & Egbert, 2008) while additionally providing ensemble uncertainty estimates not available from deterministic methods (Figure 4). The package integrates directly with ModEM (Kelbert et al., 2014), the most widely used 3D MT forward solver, by reading and writing its native file formats and wrapping it as the external forward engine for 3D VFSA inversion, ensuring interoperability with existing research workflows worldwide. The interactive 3D visualization tools (Figure 1, Figure 2) support coordinate reprojection to standard CRS (EPSG:4326, EPSG:3067) and shapefile overlays, enabling direct integration of geophysical results with geological maps in GIS platforms. The interactive model editor (Figure 3) allows polygon-based resistivity modification with depth control, supporting iterative hypothesis testing. The package's ModEM I/O and 3D visualization capabilities are actively used for interpreting crustal-scale MT surveys in Finland. The repository includes complete reproducible examples: COMMEMI synthetic benchmarks are generated natively, and the Cascadia 3D example runs end-to-end from a single script, making the package ready for adoption by the broader MT research community.

AI usage disclosure

GitHub Copilot was used as a coding assistant during development of this software and in drafting this paper. All AI-generated code and text were reviewed, tested, and verified by the author for correctness.

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